

RAG Test Document

This document is intentionally created for testing a Retrieval-Augmented Generation (RAG) pipeline. It contains factual information that can later be retrieved by a vector database and used by an LLM.

1. What is RAG?

Retrieval-Augmented Generation, commonly called RAG, is a technique that combines information retrieval with text generation. Instead of relying only on information learned during model training, a RAG system retrieves relevant information from an external knowledge source and provides that information to a language model.

2. Document Loading

A document loader reads information from a source such as a PDF, text file, web page, or database. For a PDF-based RAG system, the loader extracts the text from the PDF and converts it into a form that the application can process. Document metadata can also contain useful information such as the source file or page number.

3. Chunking

Large documents are usually divided into smaller pieces called chunks. Chunking makes retrieval more precise because the system can search for a relevant section instead of processing the entire document. A chunk size of 500 characters with an overlap of 50 characters is one possible configuration, although the best values depend on the application.

4. Embeddings

An embedding model converts text into a numerical vector. Texts with similar meanings tend to have vectors that are closer together in the embedding space. These vectors allow a system to perform semantic similarity search.

5. Vector Database

A vector database stores embeddings and allows an application to search for vectors that are similar to a user's query. Examples of vector databases and vector stores include Chroma, Pinecone, FAISS, and other systems that support vector similarity search.

6. Retrieval

When a user asks a question, the question can first be converted into an embedding. The retriever then searches the vector store and returns the most relevant document chunks. These retrieved chunks become context for the language model.

7. Generation

The language model receives the user's question together with the retrieved context. It uses that context to generate an answer. A well-designed RAG system should answer using the retrieved information rather than inventing unsupported facts.

Test Questions

Question	Expected source
What is RAG?	Section 1
Why do we use chunking?	Section 3
What does an embedding model produce?	Section 4
What is the purpose of a vector database?	Section 5
What happens during retrieval?	Section 6
What does the language model receive during generation?	Section 7